

Biochemistry paper 2 notes - Glycation chemistry

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1 Lobry de Bruyn-Alberda van Ekenstein transformation

Definition 1.1. *Base-catalyzed isomerization of reducing sugars where an aldose becomes a ketose (reversible reaction) via enediol intermediate.*

Mechanism:

1. **Deprotonation at the α -carbon:** base removes proton next to the carbonyl (product = enolate);
2. **Enediol formation:** Protonation of oxygen yields cis-enediol ($\text{HO}-\text{C}=\text{C}-\text{OH}$);
3. **Reprotonation at opposite carbon:** original aldose or ketose product (C2 carbonyl shift);
4. **Ring closure:** sugar reforms a cyclic structure with the new carbonyl position

Enediols are reactive - attack proteins, Amadori rearrangements, reducing end instability.

2 Bίlik reaction

Definition 2.1. *Mo(VI)-catalyzed epimerization of aldoses involving a 1,2-carbon shift, converting between aldoses.*

Reaction uses molybdic acid or Mo-based polyoxometalates, migration of carbon skeleton, stereochemistry determined by Mo.

Mechanism:

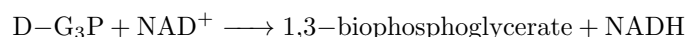
1. **Coordination of sugar to Mo(VI):** Aldehyde and adjacent hydroxyl groups bind to molybdenum center, Mo acts as electron buffer
2. **1,2-Carbon shift:** occurs at C2-C3 region, flipping stereochemistry at C2

3. **Interconversion of functional groups:** C1 aldehyde $\rightarrow$ alkoxide, C2 alkoxide $\rightarrow$ aldehyde;
4. Release of epimeric aldose.

3 Enolisation via LdB-AvE

3.1 Non-enzymatic enolisation

Example: glyceraldehyde 3-phosphate dehydrogenase (GAPDH) only oxidizes D-isomer. L-isomer is inert. Oxidation yields:



Reprotonation of enediol $\rightarrow$ enantiomer formation $\rightarrow$ racemisation. As racemisation occurs: some L-isomer transforms into D-isomer. D-isomers are oxidized by GAPDH $\rightarrow$ increase in NADH (rate of NADH change = rate of racemisation).

Reaction dihydroxyacetone phosphate (DHAP) = loss of phosphate = enol product = enediolate, catalyzed by quinuclidine bases

Enzymic enolisation: triose phosphate isomerase (TIM) mechanism: involved one catalytic base conveyed H1 of DHAP to C2 of GAP. Base = Glutamate 165. After substrate binding, loop of protein covers intermediate product and locked in place by a Tyr hydrogen.

1. DHAP binds in a precise orientation, flexible loop closes and residues bind to the active site, stabilized by Tyr hydrogen bonding, 2. Glu165 deprotonates C1 of DHAP $\rightarrow$ enediolate, 3. enediolate protonated on carbonyl oxygen $\rightarrow$ cis-enediol (stabilized via His95. Glu165 protonates C2 and produces GAP from enediol.

Isomerisation of reducing sugars by hydride shift: 1. sugar must be in its open-chain form; 2. Carbonyl at C1 polarizes the C2-H bond, yielding an oxocarbenium like TS; 3. 1,2-hydride migration (C2-C1) converting C1 aldehyde to alcohol and C2 alcohol to carbonyl resulting in ketose formation; 4. new carbonyl at C2 cyclizes yielding the ketose form.

References

- [1] SINNOTT, Michael L.; *Carbohydrate Chemistry and Biochemistry: Structure and Mechanism*; RSC Publishing (Cambridge); 2007; pp.478-492.